

DSC540Week11&12

May 22, 2025

1 Activity For Week 11 & 12

1.0.1 Activity 8.01

```
[3]: import sqlite3
      conn = sqlite3.connect("petsdb")
```

```
[5]: # a tiny function to make sure the connection is successful
      def is_opened(conn):
          try:
              conn.execute("SELECT * FROM persons LIMIT 1")
              return True
          except sqlite3.ProgrammingError as e:
              print("Connection closed {}".format(e))
              return False
      print(is_opened(conn))
```

True

```
[29]: # closing the connection for database
      conn.close()
```

```
[9]: print(is_opened(conn))
```

Connection closed Cannot operate on a closed database.
False

```
[35]: conn = sqlite3.connect("petsdb")
      c = conn.cursor()
```

```
[37]: # loop through people's age from persons database
      for ppl, age in c.execute("SELECT count(*), age FROM persons GROUP BY age"):
          print("We have {} people aged {}".format(ppl, age))
```

We have 2 people aged 5
We have 1 people aged 6
We have 1 people aged 7
We have 3 people aged 8
We have 1 people aged 9
We have 2 people aged 11

We have 3 people aged 12
We have 1 people aged 13
We have 4 people aged 14
We have 2 people aged 16
We have 2 people aged 17
We have 3 people aged 18
We have 1 people aged 19
We have 3 people aged 22
We have 2 people aged 23
We have 3 people aged 24
We have 2 people aged 25
We have 1 people aged 27
We have 1 people aged 30
We have 3 people aged 31
We have 1 people aged 32
We have 1 people aged 33
We have 2 people aged 34
We have 3 people aged 35
We have 3 people aged 36
We have 1 people aged 37
We have 2 people aged 39
We have 1 people aged 40
We have 1 people aged 42
We have 2 people aged 44
We have 2 people aged 48
We have 1 people aged 49
We have 1 people aged 50
We have 2 people aged 51
We have 2 people aged 52
We have 2 people aged 53
We have 2 people aged 54
We have 1 people aged 58
We have 1 people aged 59
We have 1 people aged 60
We have 1 people aged 61
We have 2 people aged 62
We have 1 people aged 63
We have 2 people aged 65
We have 2 people aged 66
We have 1 people aged 67
We have 3 people aged 68
We have 1 people aged 69
We have 1 people aged 70
We have 4 people aged 71
We have 1 people aged 72
We have 5 people aged 73
We have 3 people aged 74

```
[15]: for ppl, age in c.execute(
        "SELECT count(*), age FROM persons GROUP BY age ORDER BY count(*) DESC"):
        print("The highest number of people is {} and came from {} age group".
        ↪format(ppl, age))
        break
```

The highest number of people is 5 and came from 73 age group

```
[17]: res = c.execute("SELECT count(*) FROM persons WHERE last_name IS null")
        for row in res:
            print(row)
```

(60,)

```
[39]: # query to get no of people have more than one pet.
        res = c.execute("SELECT count(*) FROM (SELECT count(owner_id) FROM pets GROUP_
        ↪BY owner_id HAVING count(owner_id) >1)")
        for row in res:
            print("{} people have more than one pets".format(row[0]))
```

43 people have more than one pets

```
[21]: res = c.execute("SELECT count(*) FROM pets WHERE treatment_done=1")
        for row in res:
            print(row)
```

(36,)

```
[23]: res = c.execute("SELECT count(*) FROM pets WHERE treatment_done=1 AND pet_type_
        ↪IS NOT null")
        for row in res:
            print(row)
```

(16,)

```
[25]: res = c.execute("SELECT count(*) FROM pets JOIN persons ON pets.owner_id =_
        ↪persons.id WHERE persons.city='east port'")
        for row in res:
            print(row)
```

(49,)

```
[27]: res = c.execute("SELECT count(*) FROM pets JOIN persons ON pets.owner_id =_
        ↪persons.id WHERE persons.city='east port' AND pets.treatment_done=1")
        for row in res:
            print(row)
```

(11,)